

Takur Sai Karthik Chalamalasetty

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Summary

AI/ML Engineer with 4+ years of experience building production-grade machine learning and generative AI systems across fintech and retail supply chain domains. Specializes in LLM-powered applications, Retrieval-Augmented Generation (RAG), and embedding-based semantic search for high-volume transactional data. Strong background in scalable data pipelines, real-time inference, and anomaly detection using Spark, Kafka, and cloud platforms. Proven ability to deploy and monitor models using Kubernetes, MLflow, and Vertex AI, delivering measurable improvements in accuracy, latency, and operational efficiency.

Skills

Programming & Data Processing: Python (NumPy, Pandas, SciPy), SQL (Advanced Queries, Window Functions), Scala, Data Cleaning, Data Wrangling, Feature Engineering, Statistical Analysis

Machine Learning & Advanced Analytics: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Model Development, Hyperparameter Tuning, Model Evaluation, Time Series Forecasting, Anomaly Detection, Predictive Modeling

Generative AI & NLP: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Hugging Face Transformers, BERT, GPT, Embedding Models, Semantic Search, Text Classification, Named Entity Recognition (NER)

Data Engineering & Big Data: Apache Spark, PySpark, Databricks, Delta Lake, Apache Kafka, Snowflake, BigQuery, ETL/ELT Pipelines, Batch Processing, Streaming Processing, Feature Store Implementation, Data Versioning

MLOps & Deployment: MLflow, Kubeflow, Docker, Kubernetes, CI/CD Pipelines, Model Deployment, Model Monitoring, Experiment Tracking, Drift Detection, A/B Testing, REST APIs (FastAPI, Flask)

Cloud Platforms & Visualization: AWS (SageMaker, S3, Lambda, EC2, ECR), Azure (Azure Machine Learning, Azure Data Factory, Azure Databricks, Azure Blob Storage, Azure Kubernetes Service (AKS)), Power BI, Tableau, Matplotlib, Git, DVC, JupyterLab

Professional Experience

AI/ML Engineer, Glean

Jun 2025 – Present | Remote, USA

- Built LLM-based fraud risk classification workflows using Retrieval-Augmented Generation and prompt tuning, improving suspicious transaction detection precision by 18% across real-time digital payment processing systems.
- Developed transformer-based generative pipeline using Hugging Face to automate dispute summarization, reducing manual review workload by 22% and improving response consistency across customer support operations in fintech systems.
- Implemented embedding-driven semantic search using FAISS vector indexing, enhancing transaction categorization accuracy and enabling faster contextual retrieval across high-volume financial activity logs exceeding 2 million records monthly.
- Deployed scalable inference services using Kubernetes-based orchestration, enabling continuous processing of behavioral anomaly signals and supporting near real-time fraud monitoring across distributed cloud-hosted banking applications.
- Built model experimentation workflows with MLflow for tracking prompt variations and evaluation metrics, supporting structured A/B testing of LLM outputs and improving reliability of AI-generated financial insights across applications.
- Developed real-time credit risk scoring pipelines using Google Vertex AI, processing large-scale transactional data streams and improving risk evaluation latency by 16% across embedded financial services platforms handling millions of users.
- Built data ingestion and preprocessing pipelines to feed LLM-based workflows, enabling efficient retrieval over structured and unstructured financial datasets and reducing end-to-end inference latency by 14% across production systems.

ML Engineer, ServiceNow

Jan 2021 – Dec 2023 | Remote, India

- Developed demand forecasting models using PySpark and LightGBM, improving weekly inventory planning accuracy by 20% across multi-location warehouse networks supporting retail supply chain operations with fluctuating demand patterns.
- Built ETL pipelines using Azure Data Factory and Databricks, processing over 25 million transactional records to support consistent feature engineering and reliable downstream machine learning model training workflows.
- Implemented anomaly detection models using Scikit-learn to identify supplier delays and inconsistencies, reducing order fulfillment cycle disruptions by 15% across regional logistics and warehouse distribution operations.
- Designed batch and streaming data pipelines using Apache Spark and Kafka, enabling near real-time ingestion and transformation of supply chain data for predictive analytics and operational decision-making workflows.
- Built analytical data models using BigQuery to support supply chain performance tracking, enabling faster query execution and improving reporting efficiency for demand planning and operational decision-making teams by 18%.
- Deployed machine learning models using Docker-based containerization and Azure Machine Learning services, enabling reproducible environments and improving deployment consistency across production forecasting and analytics use cases.
- Built RESTful inference APIs using Flask to serve demand prediction outputs, enabling automated stock replenishment recommendations and improving system response times during peak transaction periods across online retail platforms.
- Designed Snowflake-based feature store to manage curated datasets and versioned features, increasing reproducibility of model training workflows and supporting experimentation across forecasting and customer segmentation use cases.

Projects

Autonomous Risk & Decision Intelligence System | FastAPI, Streamlit, SQLite, JWT, LLM APIs | (Live)

- Engineered an end-to-end AI-driven fraud risk analysis system to process unstructured case descriptions and generate automated decision outputs for real-time workflows.
- Developed FastAPI-based REST APIs with JWT authentication, enabling secure user-specific data access and scalable backend processing.
- Designed an interactive Streamlit dashboard to visualize risk scores, case trends, and decision insights, improving interpretability of model outputs.
- Implemented entity extraction (amount, date) and rule-based risk scoring logic to classify cases into risk categories and automate escalation decisions.
- Built persistent data storage using SQLite and SQLAlchemy, enabling user-level case history tracking and real-time data retrieval.
- Deployed full-stack application using Render (backend) and Streamlit Cloud (frontend), ensuring production accessibility and integration.

Automated Detection and Classification of Underwater Species | Python, YOLOv8, Deep Learning, OpenCV

- Built real-time underwater object detection and segmentation system using YOLOv8, enabling accurate multi-species identification under challenging low-visibility marine environments.
- Preprocessed and augmented multi-class underwater datasets, addressing class imbalance, noise, and poor lighting conditions to improve model generalization and robustness significantly.
- Optimized model performance using transfer learning and feature engineering, improving detection efficiency while maintaining stability across varying underwater conditions and image distortions.
- Achieved 80.82% precision, 69.35% recall, 76.86% mAP@0.5, and 85.48% segmentation accuracy, demonstrating strong performance on complex underwater datasets.

Security of Drone-Based Delivery Systems | Python, Flask, Cryptography, Simulation, Data Analysis

- Designed multi-layer security framework for drone communication, ensuring secure data transmission, delivery validation, and user authentication across distributed drone delivery network systems.
- Developed simulation integrating AES-256 encryption, SHA-256 hashing, and OTP-based verification to safeguard communication channels and prevent unauthorized access during operations.
- Generated and analyzed security logs under simulated attack scenarios, evaluating system resilience, identifying vulnerabilities, and improving threat detection accuracy across multiple attack vectors.
- Achieved 100% attack prevention with minimal overhead, maintaining under 50 milliseconds encryption latency and average delivery time of 30.6 seconds.

Education

Master of Science , <i>Lawrence Technological University</i> Computer Science	Jan 2024 – Dec 2025 MI, USA
Bachelor of Science , <i>SRM University AP</i> Computer Science	Jun 2019 – May 2023 AP, India